1. The **factorial function** $n!$ is defined as $1 \cdot 2 \cdot \cdots \cdot n$, i.e. it is $\prod_{i=1}^{n} i$. Consider the following ***incorrect*** example of proof by induction.

Claim: For all natural numbers $n \geq 1$, $n! > 2^n$.

Proof: Let us proceed by induction over n. Suppose that $n! > 2^n$ for some $n \geq 1$. We shall show that $(n+1)! > 2^{n+1}$.. By definition, $(n+1)! = (n+1) \cdot 2^n \geq 2^{n+1}$. Therefore, $(n+1)! > 2^{n+1}$; thus completing our proof by induction.

Find an example $n \geq 1$ such that $n! > 2^n$ and explain why the proof is incorrect.

Let $n = 1$. Following, $n! = 1, 2^n = 2$. $n! \not> 2^n$. Therefore, this proof is incorrect. This is because the proof hasn't considered the base case of $n = 1$.

2. Prove the following claim.

Claim: For all integers $n \geq 4$, we have $n! > 2^n$.

Consider the base case $n = 4$. Following, $n! = 24$ and $2^n = 16$. $n! > 16$.

Let $F(n)$ be the factorial function, and $G(n)$ be 2^n.

Now consider the case $F(n+1) = (n+1) \cdot F(n)$.

Also consider the case $G(n+1) = 2 \cdot 2^n$.

Assume $F(n+1) > G(n+1)$.

$(n+1) \cdot n! > 2 \cdot 2^n$.

$\frac{(n+1) \cdot n!}{2} > 2^n$. IH

Since $n! > 2^n$, we have proved it.

3. Consider the following proof by induction.

Claim: For all natural numbers n, the sum of the first n odd numbers is n^2.

$$\sum_{i=0}^{n-1} (2i+1) = n^2$$

Let $S(n) = \sum_{i=0}^{n-1} (2i+1)$ and

$R(n) = n^2$

Consider the base cases $S(1), R(1)$.

$S(1) = R(1) = 1$

Now consider the cases $S(n+1), R(n+1)$.

$S(n+1) = (2n+1) + \sum_{i=0}^{n-1} (2i+1)$

Using IH and assuming $S(n) = R(n)$:

$S(n+1) = (2n+1) + R(n)$

$S(n+1) = (2n+1) + n^2$

$S(n+1) = n^2 + 2n + 1$

$S(n+1) = (n+1)^2$, which is just $R(n+1)$.

4. For each of the following claims, write a proof by induction:

a. $\sum_{i=0}^{n} 2^i = 2^{n+1} - 1$;

Let:

$S(n) = \sum_{i=0}^{n} 2^i$ and

$R(n) = 2^{n+1} - 1$

Consider the base case $S(0), R(0)$:

$S(0) = R(0) = 1$

Consider $S(n+1), R(n+1)$

$S(n+1) = 2^{n+1} + \sum_{i=0}^{n} 2^i$

Using IH, and assuming $S(n) = R(n)$:

$S(n+1) = 2^{n+1} + (2^{n+1} - 1)$

$S(n+1) = 2(2^{n+1}) - 1$

$S(n+1) = 2^{n+2} - 1$, which is just $R(n+1)$.

b. $\sum_{i=0}^{n} r^i = \frac{1-r^{n+1}}{1-r}, r \neq 0, r \neq 1$;

Let:

$S(n) = \sum_{i=0}^{n} r^i$ and

$R(n) = \frac{1-r^{n+1}}{1-r}$

Consider the base case $S(0), R(0)$:

$S(0) = 1$

$R(0) = \frac{1-r}{1-r} = 1$

$S(0) = R(0)$

Consider the cases $S(n+1), R(n+1)$:

$S(n+1) = r^{n+1} + \sum_{i=0}^{n} r^i$

Using IH, and assuming $S(n) = R(n)$:

$S(n+1) = r^{n+1} + R(n)$

$S(n+1) = r^{n+1} + \frac{1-r^{n+1}}{1-r}$

$S(n+1) = \frac{r^{n+1}(1-r)}{1-r} + \frac{1-r^{n+1}}{1-r}$

$S(n+1) = \frac{r^{n+1}-r^{n+2}}{1-r} + \frac{1-r^{n+1}}{1-r}$

$S(n+1) = \frac{1-r^{n+2}}{1-r}$, which is just $R(n+1)$

5. Prove that $n^3 - n$ is divisible by 3 for all $n \in \mathbb{N}$.

Let $S(n) = 3 \mid n^3 - n$

Consider the base case $S(1)$:

$S(0) = \top$

$S(n) \Rightarrow S(n+1)$

Consider the case $S(n+1)$:

$S(n+1) = 3 \mid (n+1)^3 - (n+1)$

$S(n+1) = 3 \mid n^3 + 3n^2 + 3n + 1 - n - 1$

$S(n+1) = 3 \mid 3n^2 + 3n + 1 + n^3 - n - 1$

Using IH, and assuming $S(n+1)$

$S(n+1) = 3 \mid 3n^2 + 3n + 1 + S(n) - 1$

$S(n+1) = 3 \mid (3n^2 + 3n) + (n^3 - n)$

$S(n+1) = 3 \mid 3(n^2 + n) + 3b$

$S(n+1) = 3 \mid 3a + 3b$

$S(n+1) = 3 \mid 3(a + b)$

$S(n+1) = \top$